

In this paper, every Ring is Commutative Ring with $1 \neq 0$.

Thm. Let $A_1, \dots, A_k$ be Ideals in R . Define a map

$$\mathcal{Q}: R \rightarrow R/A_1 \times \dots \times R/A_k: r \mapsto (r+A_1, \dots, r+A_k)$$

is a Ring Homomorphism with $\ker \mathcal{Q} = A_1 \cap A_2 \cap \dots \cap A_k$.

If for each $1 \leq i < j \leq k$, $A_i + A_j = R$ (comaximal), then $\mathcal{Q}$ is onto and

$$\ker \mathcal{Q} = A_1 \cap \dots \cap A_k = A_1 A_2 \dots A_k.$$

Therefore, $R/(A_1 \dots A_k) = R/(A_1 \cap \dots \cap A_k) \cong R/A_1 \times \dots \times R/A_k$.

Proof. First, prove this for $k=2$. The map

$$\mathcal{Q}: R \rightarrow R/A_1 \times R/A_2: r \mapsto (r+A_1, r+A_2)$$

is just the natural projection for $R/A_1, R/A_2$ elementwise, so ring-homomorphism.

Moreover,

$$\begin{aligned} \mathcal{Q}(r) = 0 &\Leftrightarrow r+A_1 = A_1 \text{ and } r+A_2 = A_2 \\ &\Leftrightarrow r \in A_1 \text{ and } r \in A_2 \end{aligned}$$

so $\ker \mathcal{Q} = A_1 \cap A_2$. Now, suppose that $A_1 + A_2 = R$. Since $1 \in R = A_1 + A_2$,
 $x+y=1$ for some $x \in A_1$ and $y \in A_2$.

Therefore, $\mathcal{Q}(x) = \mathcal{Q}(1-y) = (x+A_1, 1-y+A_2) = (0+A_1, 1+A_2)$.

And similarly $\mathcal{Q}(y) = (1, 0)$. Now, given $(r_1+A_1, r_2+A_2) \in R/A_1 \times R/A_2$,

$$\begin{aligned} \mathcal{Q}(r_2x + r_1y) &= \mathcal{Q}(r_2)\mathcal{Q}(x) + \mathcal{Q}(r_1)\mathcal{Q}(y) \\ &= \mathcal{Q}(r_2)(\bar{0}, 1) + \mathcal{Q}(r_1)(1, \bar{0}) \\ &= (\bar{0}, \bar{r}_2) + (\bar{r}_1, \bar{0}) = (r_1+A_1, r_2+A_2). \end{aligned}$$

Thus $\mathcal{Q}$ is onto. And $A_1 A_2 \subseteq A_1 \cap A_2$, because: for $a_i \in A_1, b_i \in A_2$,

$$a_1 b_1 + \dots + a_n b_n \in A_1 \text{ and } A_2 \text{ since } A_1, A_2 \text{ are Ideal.}$$

Meanwhile, if $A_1 + A_2 = R$, for any $c \in A_1 + A_2$,

$$c = c \cdot 1 = c(x+y) = cx + cy \in A_1 + A_2.$$

So proved.

Thm. Let R be a ring with identity $1 \neq 0$.

Let $E \in R$ be an idempotent in R s.t. $Er = rE$ for all $r \in R$.

Then, $RE = R(1-E)$ are two-sided ideals of R , and that

$$R \cong RE \times R(1-E).$$

Moreover, E and $1-E$ are identities for RE and $R(1-E)$ respectively.
Proof.

$$DE_1 \cdots E_n =$$